Robotics



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Lecture # 4

Sensors for Robotic Systems II

Encoders

Encoders

Encoders

- The term **encoder** is used for a device that **provides a digital output** as a result of **angular or linear displacement**.
- An **increment** encoder **detects changes** in **angular or linear displacement** from some datum position,
- while an **absolute** encoder **gives the actual angular or linear position**.
- Encoders are **specific sensors** that **generate square pulses** in response to a wide range of motion tasks.

Encoders

Encoders

- In many manufacturing and production processes, there is a **need** for **precise shaft rotation positioning or linear motion position measuring.**
- For **this purpose**, there are **shaft or hollow shaft encoders** that respond to rotation (**rotary encoders**) and **linear encoders** that respond to a linear movement along a straight-line motion.
- As electromechanical devices, encoders are able to **follow the detected** motion **through mechanical coupling** and provide a motion control system with **feedback information**, **concerning position or distance**, **rotation speed** or **velocity**, and **direction**.
- **Linear** and **rotary** encoders are both available with **two main types** of **output** interpretation: the **absolute encoder** and the **incremental encoder**.

Incremental Encoder

Encoders

Incremental Encoder

- The **figure shows** the **basic form** of an **incremental** encoder for the measurement of angular displacement.

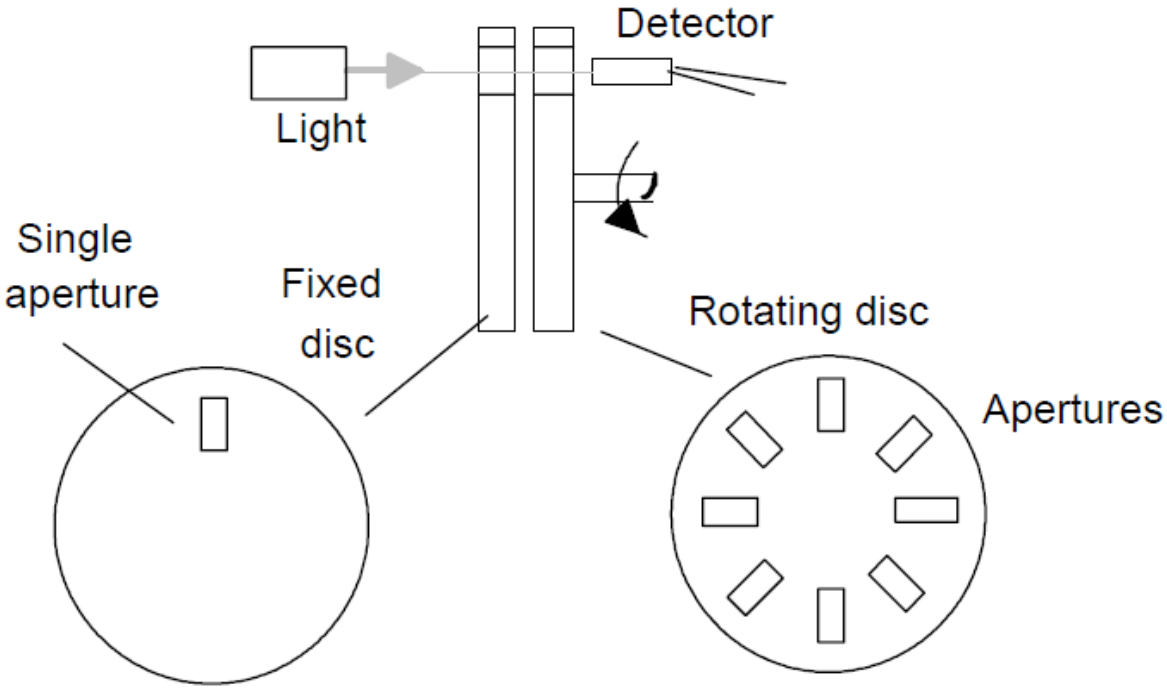


Figure: Basic form of an incremental encoder

Encoders

Incremental Encoder

- A **beam of light**, from perhaps a **light-emitting diode** (**LED**), **passes** through **slots** in a **disc** and is detected by a **light sensor**, e.g. a **photodiode** or **phototransistor**.
- When the **disc rotates**, the **light** beam is **alternately transmitted** and **stopped** and so **a pulsed output is produced** from the light sensor.
- The **number** of pulses **is proportional to** the **angle** through which the disc has rotated, the **resolution** being **proportional to** the **number of slots** on a disc.
- With **60 slots** then, since **one revolution** is a rotation of **360**˚, a movement from **one slot** to the next is a rotation of **6**˚.
- By using **offset slots** it is possible to have **over a thousand** slots for one revolution and so **much higher resolution**.

Encoders

Incremental Encoder

- An **incremental** encoder may have **one or more outputs** of pulses called **“channels”.**
- A **single channel output can give only the position**, and hence is **not used** in applications where **direction of movement is required**.
- A **two-channel encoder** produces **two pulse trains** that are **90° out of phase** with each other.
- Then, the **direction of movement** can be **determined** through the **phase relationship between them**.

Encoders

Incremental Encoder

- **Depending on** the number of the **rising and falling points** of the **two pulse trains** that are counted, the **counting resolution** may be **single, double, or quadruple.**
- **Single** resolution corresponds to **rising edges** of **one** channel only.
- **Double resolution** is obtained when **both rising and falling edges** of **one** channel only are counted.
- **Quadruple resolution**, the **most precise** motion detection, is **achieved** when **both rising and falling edges** are **counted** from **both** channels.

Encoders

Incremental Encoder

- Some incremental encoders can also **include** an **additional channel** known as the **"0-index pulse"** or **"Z channel",** producing **one pulse per rotation** for a **rotary encoder or at specific**, **precisely-known positions for a linear encoder.**
- Since, the **incremental encoders don't have** a **constant reference** point (as absolute encoders do), the **0-index signal** can be used for the **homing procedure at startup**.
- **For example**, the **operation** of an incremental encoder, shown in the figure, for the case of an encoder with **dual channel** and z**ero index output.**

Encoders

Incremental Encoder

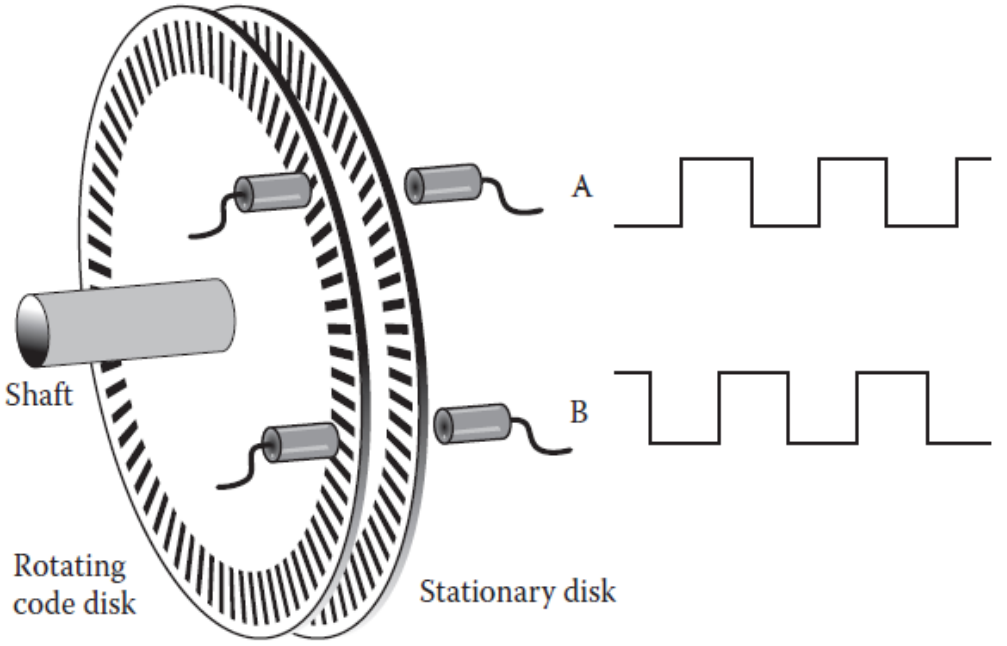


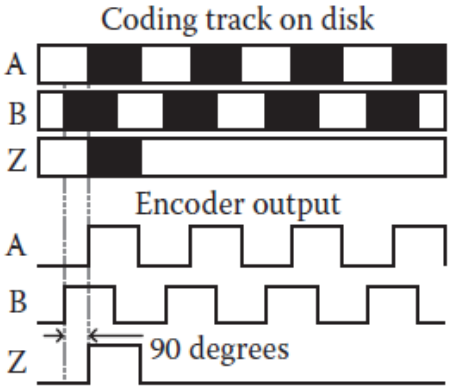
Figure: Pulse trains of a two-channel encoder and zero index for resulting coding data.

Coding for clockwise rotation

Phase	A	B
1	0	0
2	0	1
3	1	1
4	1	0

Coding for counterclockwise rotation

Phase	A	B
1	1	0
2	1	1
3	0	1
4	0	0



Encoders

Incremental Encoder

- From this figure, the **resulting pulse trains** for the channels **A and B** are also **represented**, that are also **different for different rotation directions** and the resulting **coding** for the **clockwise** and the **counter-clockwise operation.**
- **Based on** the **accuracy** of the encoder and the **number** of district **pulses per rotation**, the **rotation achieved can be measured**.
- The bottom part of this figure also **indicates a 2D representation** of the pulse trains, and **a zero index** in the case of a **quadrature encoder.**

Absolute Encoder

Encoders

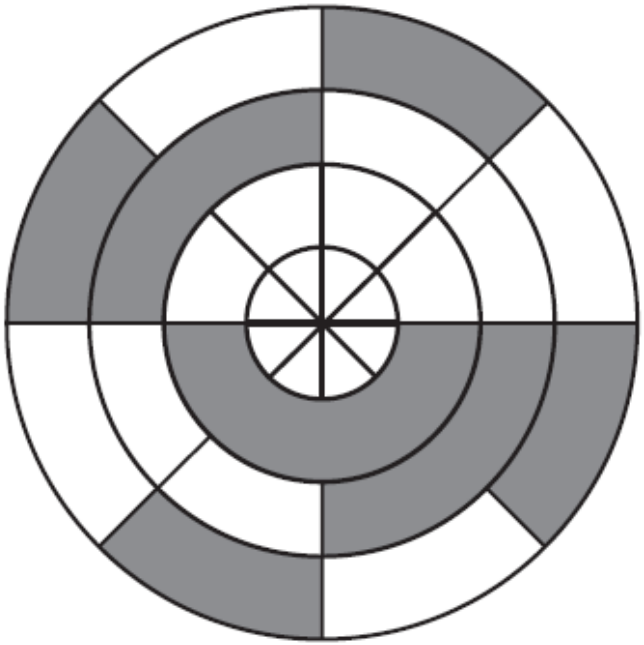
Absolute Encoder

- The **absolute** encoder **differs from** the **incremental** encoder in **having a pattern of slots which uniquely defines each angular position.**
- With the form shown in the **figure**, the **rotating disc** has **four concentric circles of slots** and **four sensors to detect the light pulses.**
- The **slots** are **arranged** in such a way that the **sequential output** from the **sensors** is a **number in the binary code**, each such number corresponding to a **particular angular position**.
- On other words, the **output** of incremental encoders **generates** a **series** of **pulses proportional** to the **rotation** of the shaft (rotary encoder) or **distance** traveled (linear encoder).

Encoders

Absolute Encoder

- This operation is presented in the figure for the case of an encoder with **three digits of accuracy.**



Standard binary encoding				
Sector	Contact 1	Contact 2	Contact 3	Angle
0	OFF	OFF	OFF	0°–45°
1	OFF	OFF	ON	45°–90°
2	OFF	ON	OFF	90°–135°
3	OFF	ON	ON	135°–180°
4	ON	OFF	OFF	180°–225°
5	ON	OFF	ON	225°–270°
6	ON	ON	OFF	270°–315°
7	ON	ON	ON	315°–360°

Figure: Binary encoded output for a full rotation of an encoder.

Encoders

Absolute Encoder

- In the **left part** of this figure the **encoding** of the encoder is **presented**, **marked with black and white regions** that can be **detected** by proper **photodiodes**, while in the **right part**, the **resulting encoding** and the corresponding **rotation measurements** are **indicated**.
- For **example**, in the **case** of an **"ON-OFF-OFF"** measurement that corresponds **to line 4**, a rotation between **180° and 225°** has been achieved.
- If more **accuracy** in the positioning task is **needed**, **alternative** types of encoders with **more digits** of accuracy should be **utilized**.

Encoders

Absolute Encoder

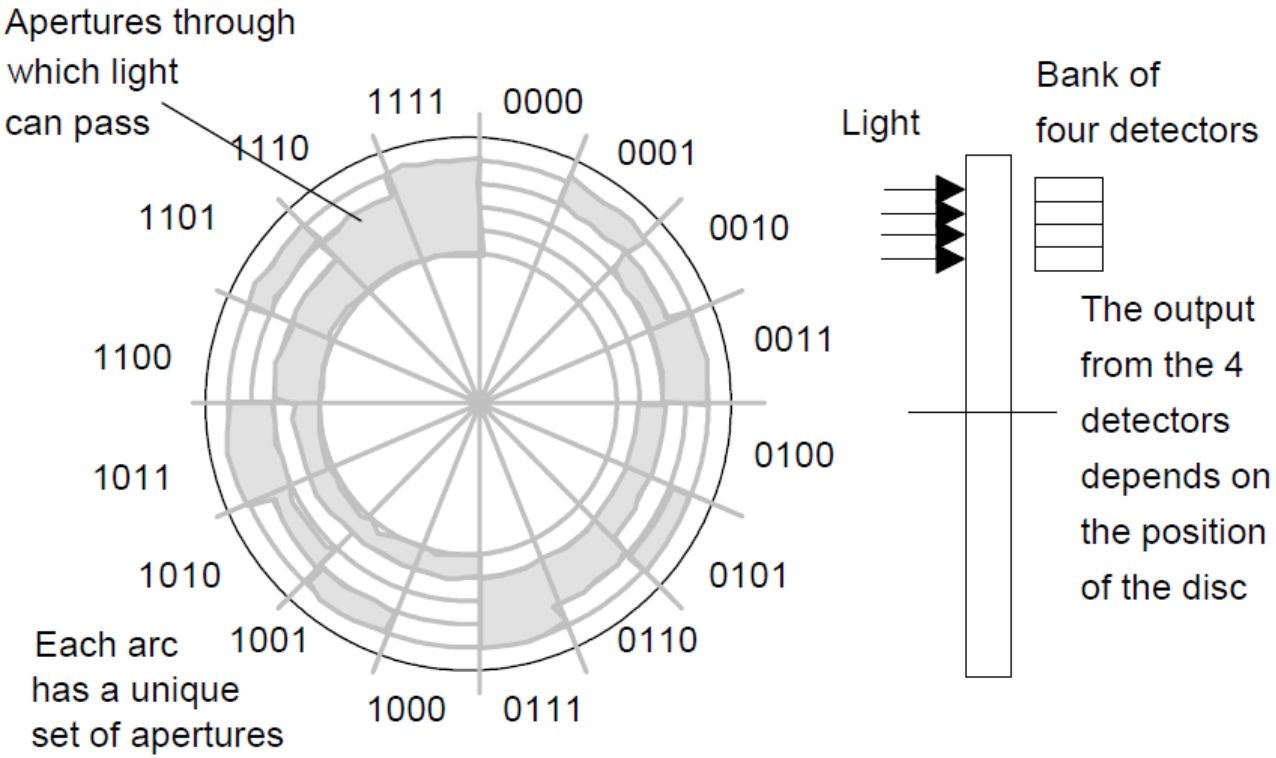


Figure: The rotating wheel of the absolute encoder. **Note** that though the normal form of binary code is shown in the figure, in practice a modified form of binary code called the **Gray code** is generally used. This code, **unlike normal binary**, has **only one bit** changing in moving from one number to the next. Thus we have the sequence 0000, 0001, 0011, 0010, 0110, 0111, 0101, 0100, 1100, 1101, 1111.

Encoders

Absolute Encoder

- With **4 tracks** there will be **4 bits** and so the **number of positions** that can be detected is **$2^4 = 16$**, i.e. a **resolution of 360/16 = 22.5°.**
- Typical encoders **tend to have up to 10 or 12 tracks.**
- The number of **bits** in the **binary number** will be **equal to** the number of **tracks**.
- Thus with **10 tracks** there will be **10 bits** and so the **number of positions** that can be detected is **2^{10}, i.e. 1024,** a **resolution of 360/1024 = 0.35°.**

Encoders

Absolute Encoder

- The **output** of absolute encoders is a **multi-bit digital code** that **indicates** the **actual** position of the moving mechanical object directly.
- This **means** that an **absolute** encoder **has a reference position**.
- By counting the number of **pulses** per a desired period of motion and **based** on the **resolution** of the encoder (i.e., the **number** of **pulses** per **revolution** or **millimeters** per **unit travel** given by the **manufacturer) the** angular or linear position can be **determined**.
- Even in the event of a **power outage**, it **has the ability** to **maintain** a **record** of its **absolute position.**
- After **restarting**, the **motion system** can **resume** motion **immediately without** the need for **rehoming**.

Encoders

Encoders

- **Absolute** and **incremental** encoders can base their operation on **optical**, **magnetic**, or **laser technology**.
- It is **noticeable** that encoders **cannot be used** with a **conventional automation circuit**, **except if** a specific **electronic unit accepts** the pulses and produces some **digital contact-type outputs** inserted in the automation circuit.
- Such **applications** are met in **robotics** for **controlling and positioning joints**, in computer numerical control **(CNC) machines**, **drilling machines**, **assembly machines**, **cutting machines** with a measuring **wheel-encoder**, and many **others**.

Sonar Sensors

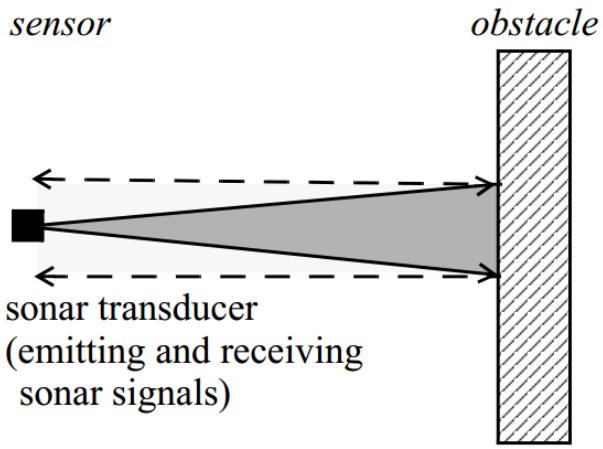
Sonar Sensors

Sonar sensors

- In the past, most robots have been equipped with sonar sensors (often Polaroid sensors).
- Because of the relatively narrow cone of these sensors, a typical configuration to cover the whole circumference of a round robot required 24 sensors, mapping about 15° each.

Sonar Sensors

- **Sonar sensors use the following principle:**
  - a short acoustic signal of about 1ms at an **ultrasonic** frequency of 50kHz to 250kHz is emitted and the time is measured from signal emission until the echo returns to the sensor.
  - The measured **time-of-flight** is proportional to **twice** the distance of the nearest obstacle in the sensor cone. If **no signal** is received within a certain time limit, then no obstacle is detected within the corresponding distance.
  - Measurements are repeated about **20 times per second**, which gives this sensor its typical clicking sound (see Figure).



Sonar Sensors

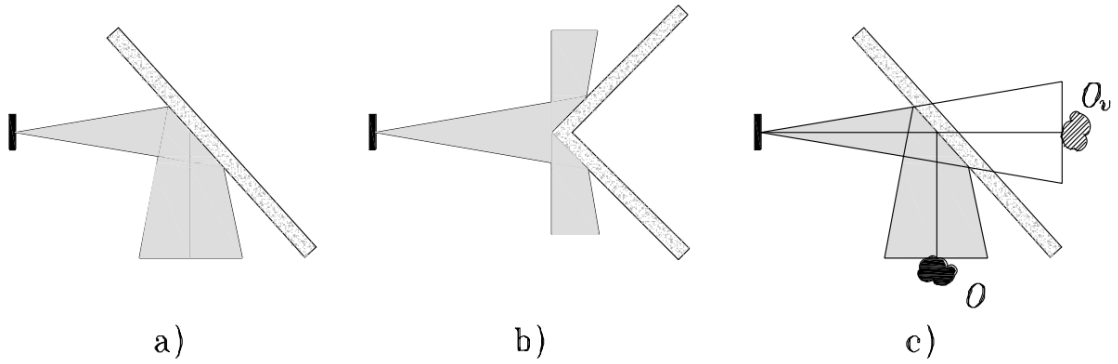
- Sonar sensors have a number of **disadvantages** but are also a very powerful sensor system.
- The most significant **problems** of sonar sensors are **reflections and interference.**
  - When the acoustic signal is reflected, for example off a wall at **a certain angle**, then an obstacle seems to be further away than the actual wall that reflected the signal.
  - Interference occurs when **several sonar sensors** are operated at once (among the **24 sensors** of one robot, or among several independent robots). Here, it can happen that the **acoustic signal** from one sensor is being picked up by another sensor, resulting in **incorrectly** assuming a closer than actual obstacle.
  - **Coded sonar** signals can be used to prevent this, for example using pseudo random codes.

Sonar Sensors

- **Piezoelectric** and **electrostatic transducers** are the two major types available that operate in air and can in principle operate both as a transmitter and receiver.
- **Accuracy** of ultrasound range sensors depends on the **features of the transducer** and the excitation/detection circuitry, as well as on the reflective properties of the surfaces hit by the acoustic waves.

Sonar Sensors

- **Smooth surfaces**, i.e., those characterized by irregularities of comparable size to that of the wavelength corresponding to the employed frequency, may produce a **non-detectable echo** at the sensor (Fig. a,b) if the incident angle of the ultrasound beam exceeds a given **critical angle** which **depends on the operational frequency** and the reflective material.
- When operating in **complex environments**, such mirror reflections may give rise to multiple reflections, thus causing range measurement **errors or false detection** (Fig. c).



Reflector models on smooth surfaces: a) non-detected plane. b) non-detected corner. c) plane with false detection (O real object, O*v* virtual object detected)

Laser Rangefinder and Laser Scanner

Laser Rangefinder and Laser Scanner

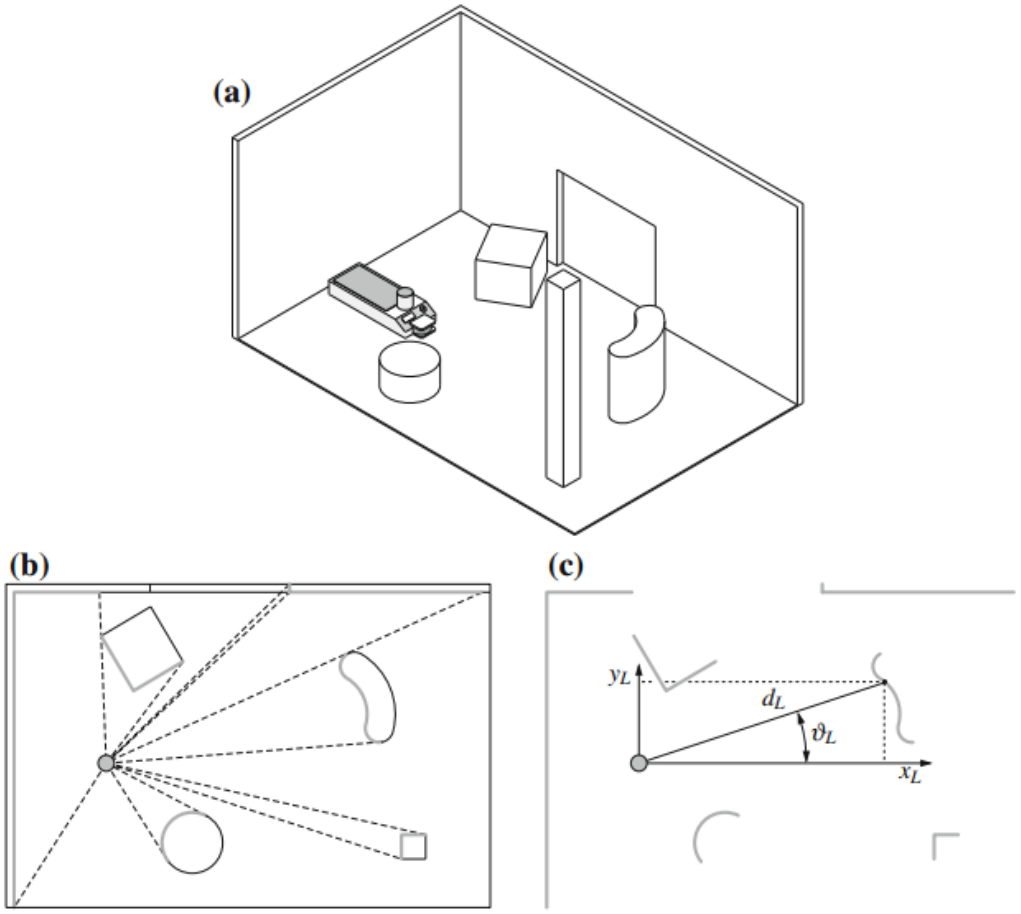
Laser Rangefinder and Laser Scanner

- A laser rangefinder uses a **laser beam** to determine the **distance** to an object.
- The most common form of laser rangefinder operates on **the time-of-flight principle**.
- Distance can be determined by **measuring** the time it takes for the laser pulse to return to the sensor and it requires precise time measuring. With the **speed of light known** and an accurate measurement of the time taken, the distance can be calculated.
- Another possibility is to compute **light-wave phase shift** by analyzing the incoming light and **comparing it to a reference signal**. The most accurate method to measure **changes in distance** rather than absolute distances is interferometry.

Laser Rangefinder and Laser Scanner

- The laser rangefinder measures **distance to one object at a time**. Therefore, it is a one-dimensional sensor.
- It can be used to generate a map of the environment, for **path planning**, and **avoidance of obstacles**.
- A three-dimensional (**3-D**) scanner enables scanning of a complete 3-D space and collecting a 3-D point cloud data by using another degree of freedom at the rotating assembly.
- These 3-D laser-scanners are typically termed **LiDAR** (Light Detection And Ranging) and are often used in **autonomous vehicles** for scanning the environment.

Laser Rangefinder and Laser Scanner



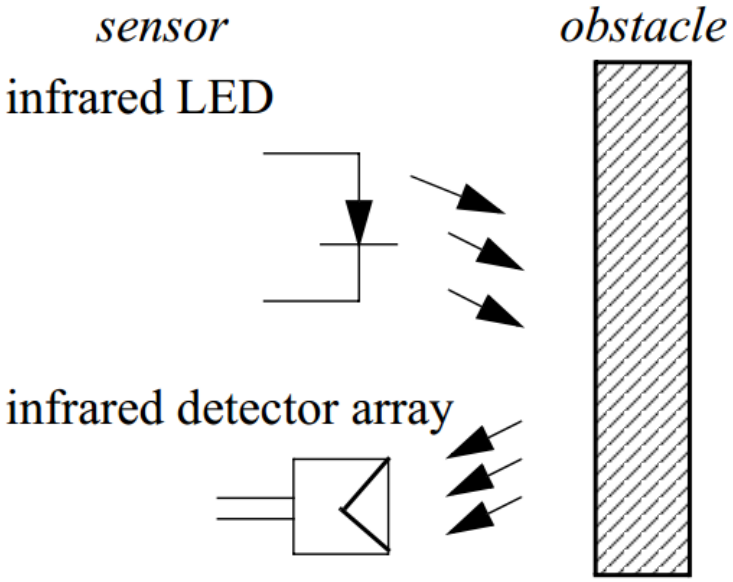
Laser scanner used to create a map of environment: a) environment, b) scanning, and c) map

Infrared Sensors

Infrared Sensors

Infrared sensors

- Infrared (IR) distance sensors **do not follow** the same principle as sonar sensors, since the time-of-flight for a photon would be much **too short** to measure with a simple and cheap sensor arrangement.
- Instead, these systems typically use a **pulsed infrared LED** at about 40kHz together with a detection array (see Figure).



Infrared Sensors

- The **angle** under which the **reflected beam** is received **changes** according to the distance to the object and therefore can be used as a measure of the distance.
- The **wavelength** used is invisible to the human eye and it can be transformed to visible light either by **IR detector cards** or by **recording the light beam** with **an IR-sensitive camera**.

Tachometer

Tachometer

Tachometer

- The signal of the joint **velocity** can be obtained by **numerical differentiation** of the **position** signal.
- Nevertheless, **direct** measurement of the joint velocity with the help of a tachometer is often used in robotics.
- The **reason** is the **noise** introduced by numerical differentiation, which can greatly affect the quality of the robot control.

Tachometer

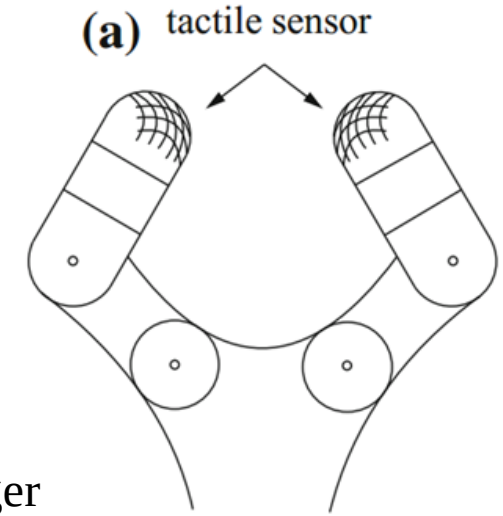
- Tachometers can be divided into: (1) direct current (**DC**) and (2) alternate current (**AC**) tachometers.
- In robotics it is generally the simpler **DC** tachometers that are used.
- The **working principle** is based on a **DC generator** whose magnetic field is provided by **permanent magnets**.
- As the magnetic field is **constant**, the tachometer output **voltage** is **proportional to the angular velocity** of the rotor.
- Because commutators are used in the **DC** tachometers, a slight ripple appears in the output voltage, which cannot be entirely filtered out.
- This deficiency, together with other imperfections, is avoided by using **AC** tachometers.

Tactile Sensor

Tactile Sensor

Tactile Sensor

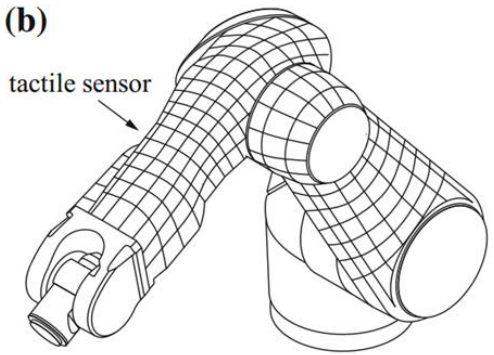
- Robots can collect information about the environment also through **touch**.
- In order to increase robot manipulation **capabilities**, tactile sensors can be used in robotic **fingers** as shown in Fig. a.
- The sensor provides data about **contact force distribution** between the finger and the manipulated **object**.



Tactile sensor used in robot finger

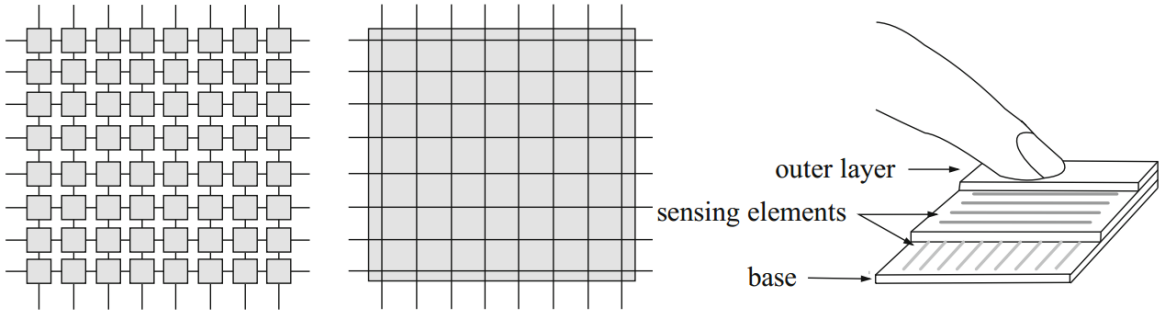
Tactile Sensor

- To increase robot safety (e.g., when working with a human), tactile sensors can be used as an **artificial robot skin** that enables the robot to **sense** contacts with objects in the environment.



Tactile sensor used as robot skin

- Tactile sensing is based on **an array of touch sensors** as shown in Figure.



A model of a tactile sensor

Tactile Sensor

The following sensing **principles** can be implemented in the array:

- **deformation-based sensors**: material surface deforms (**changes length**), when it is subjected to external forces; deformation is **converted** to **electrical** signals with **strain gauges** connected in a Wheatstone bridge;
- **resistive sensors:** electrical **resistance** changes with pressure of a material placed between two electrodes;
- **capacitive sensors:** sensing element is a **capacitor** whose capacitance **changes** with the applied **force**; force can **produce** either a **change** in the **distance** between capacitor plates or its area;

Tactile Sensor

- **optical sensors:** sensing is typically **based on light intensity measurement**; the intensity of the received light is a **function** of displacement and hence of the **applied force**;
- **magnetic sensors:** changes of **magnetic flux** density or **magnetic coupling** between circuits are the **most widely used principles** in magnetic tactile sensing;
- **mechanical sensors:** sensing elements are mechanical micro-switches with on and off states.

Thank you for your attention !

Good Luck !

